

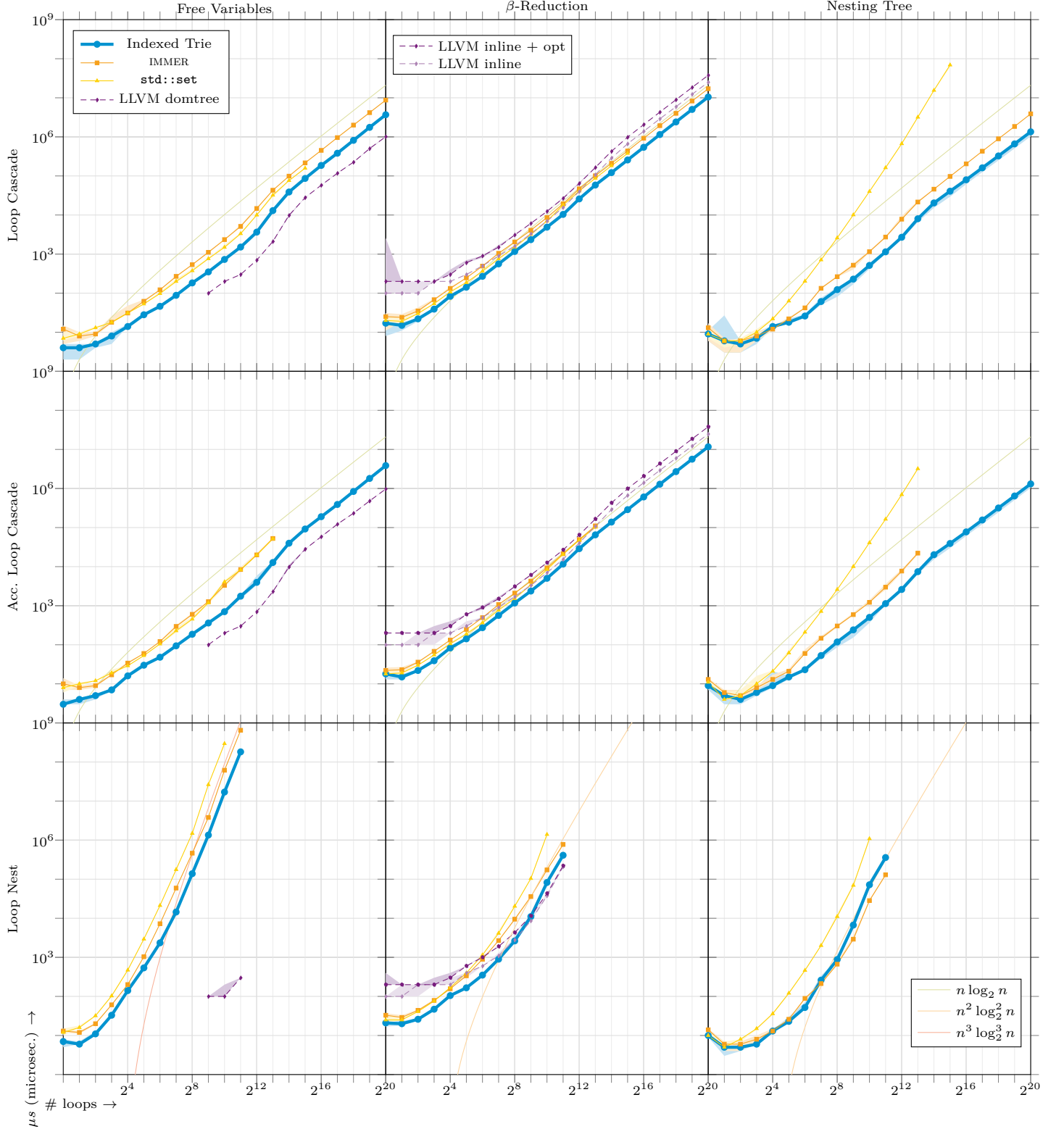


Figure 1: Performance (in μs , lower is better) of three operations—free variable computation, β -reduction, and nesting tree computation—on three synthetic programs with increasing numbers of loops. The plots compare three different set implementations: the indexed trie, IMMER, and `std::set`. Additionally, free variable computation is compared against LLVM’s dominance analysis; β -reduction is compared against LLVM’s inliner, both with and without optimizations that simulate MIMIR’s on-the-fly optimizations. For reference, $10^3 \mu\text{s} = 1 \text{ ms}$, $10^6 \mu\text{s} = 1 \text{ s}$.